
CAPSTONE PROJECT

ENERGY COMPANY

Travel & Fleet Management Process Improvement



Capstone Project for Energy Company

Task 1: Gap Analysis & Root Cause Analysis

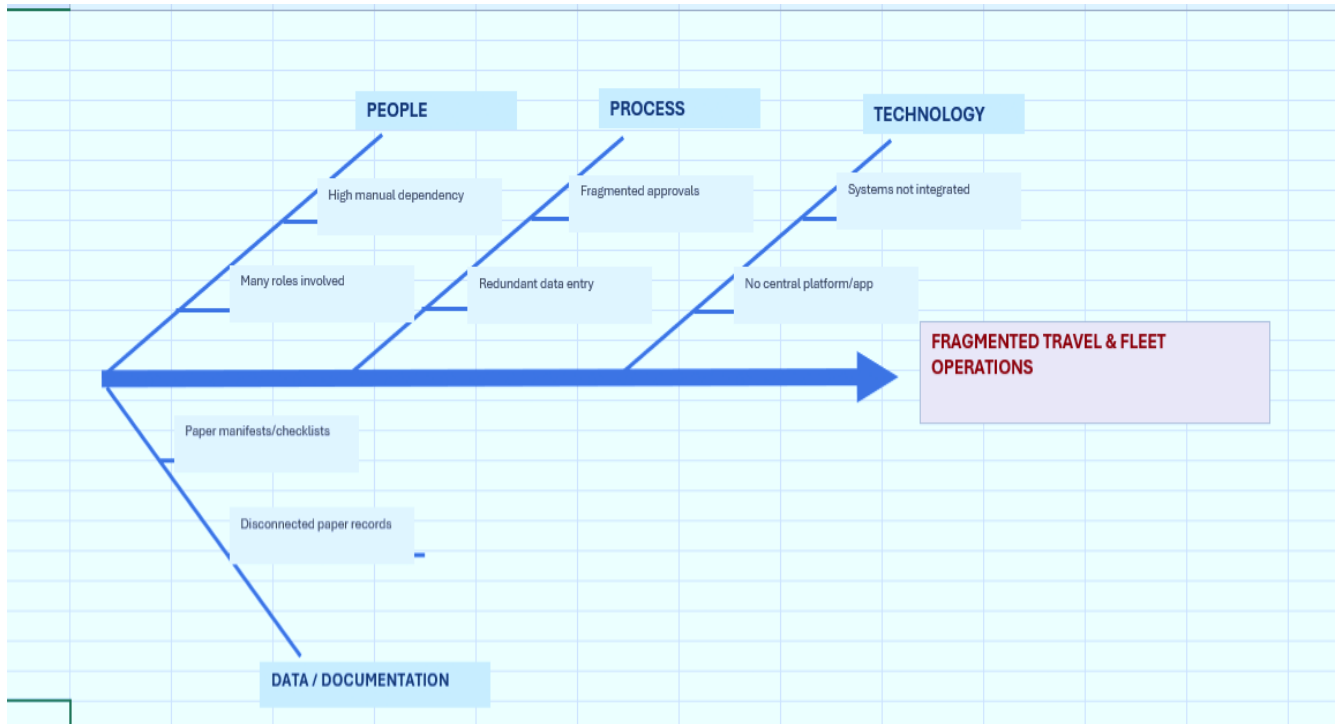
1. Gap Analysis Matrix

Business Area / Process	Current State (AS-IS)	Desired Future State (TO-BE)	Identified Gap	Recommended Action
Travel & Fleet Request Integration	Travel requests are submitted via portal, but supplemental journey details are manually sent to Logistics.	Automated data sync between Travel Portal and Fleet Management Platform.	Lack of API/system integration causing duplicate entries & manual workload.	Build direct integration/API between Travel Portal and Fleet Platform.
Security Risk Assessment	Manual risk evaluation using disjointed physical/digital forms and multi-party sign offs.	Automated security workflow routing, risk level checks, and digital approvals.	Disjointed records; slow manual sign-off tracking.	Implement digital security assessment forms with rule-based auto-approval/escalation.
Driver Assignment & Planning	Fleet Manager manually coordinates driver availability, vehicle suitability, and scheduling.	System-automated driver assignment based on real-time availability and license type.	High administrative overhead and delay in manual fleet planning.	Deploy automated dispatch engine with driver roster & vehicle availability matching.
Vehicle Inspection & Defect	Drivers complete manual	Mobile app pre-trip digital	No centralized inspection	Introduce digital pre-trip inspection

Business Area / Process	Current State (AS-IS)	Desired Future State (TO-BE)	Identified Gap	Recommended Action
Management	paper checklists; defects follow separate manual procedures.	checklist with automated defect flagging and maintenance alerts.	database or real-time defect reporting.	module on mobile app with automatic maintenance integration.
Passenger Manifest & Verification	Paper manifest signed manually before/after trip and physically submitted to Fleet Manager.	Digital QR-code/mobile check-in with real-time passenger manifest logging.	Reliance on physical documentation and manual filing.	Implement digital passenger check-in/out via mobile app.
End-to-End Visibility & Monitoring	Disconnected workflows prevent unified journey tracking across stages.	Centralized dashboard tracking real-time status from request to completion.	Operational blind spots; lack of real-time audit trail.	Implement live journey dashboard with integrated GPS tracking telemetry.

2. Root Cause Analysis (Fishbone / Ishikawa Categories)

- **People:** High manual dependency across employees, line managers, security, fleet managers, and drivers.
- **Process:** Fragmented, multi-step approval workflows requiring redundant manual data entries.
- **Technology:** Lack of integration between Travel Portal and Fleet Systems; absence of a centralized fleet platform and mobile app driver tool.
- **Data / Documentation:** Reliance on physical paper manifests, manual inspection checklists, and disconnected paper records.



Task 2 & Task 3: AS-IS and TO-BE Process Maps

Task 2: AS-IS Process Flow Breakdown

Presentation Note – Task 2 AS-IS Process Map

This diagram shows the current travel and fleet coordination process. Requests, approvals, inspections, passenger manifests, and completion records are still handled through a mix of portal inputs, manual forms, paper documents, and separate hand-offs.

Key message: The AS-IS process has several manual touchpoints and decision delays, especially in security approval, vehicle checks, passenger manifest control, and final paperwork submission.

Geometric Shapes Legend for Process Mapping:

Shape	Meaning	Task 2 Example
Oval / Terminator	Start or end point of the process	Trip Cancelled / End; Journey Complete & Submit Paperwork
Rectangle / Process	Activity, task, or operational step	Employee: Travel Portal Request; Fleet Manager Assigns Driver
Diamond / Decision	Decision point with Yes/No outcomes	Is Approved?; Security Cleared?; Vehicle Fit?

Arrow / Connector	Direction of flow between steps	Shows movement from request submission through approvals, planning, journey execution, and completion
Document Symbol	Paper-based input, form, or record	Manual Journey Form to Logistics; Paper Passenger Manifest; Submit Paperwork
Swimlane / Role Band	Groups activities by responsible role or department	Employee, Line Manager, Security Team, Fleet Manager, Driver

Task 3: TO-BE Process Flow Breakdown

Presentation Note – Task 3 TO-BE Process Map

This diagram shows the proposed future-state process, where the travel portal, fleet management platform, digital approvals, driver app, maintenance system, and monitoring dashboard are integrated into one streamlined workflow.

Key message: The TO-BE process reduces manual administration by automating approvals, dispatching, inspections, defect escalation, passenger check-in, journey tracking, and record archiving.

Task 4: Business Requirements Document (BRD)

BUSINESS REQUIREMENT DOCUMENT – FLEET MANAGEMENT PLATFORM

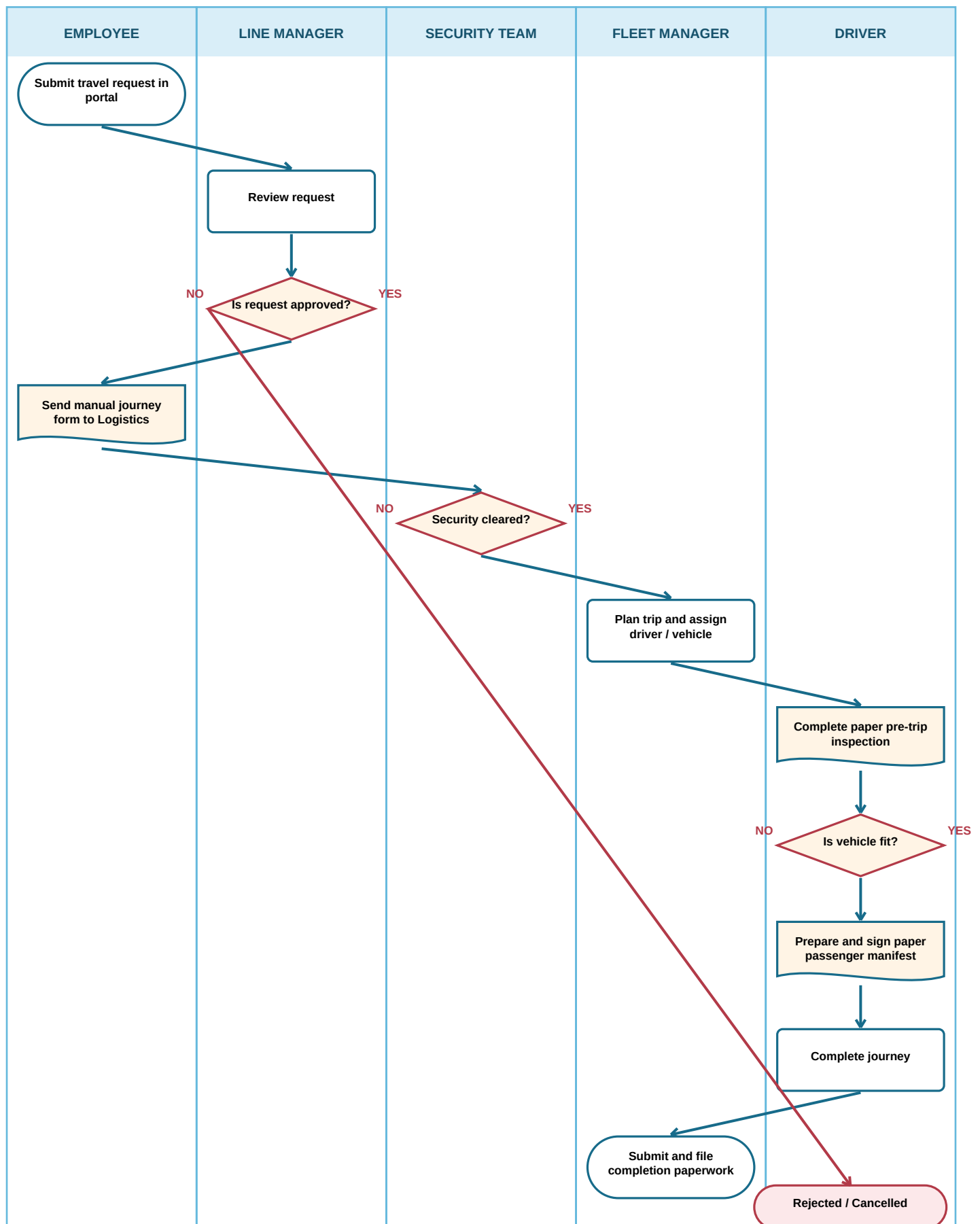
Document Ref.		BRD-FMP-2026-001	
Document Owner		Fleet and Logistics Department	
Project Deployment Date		01-12-2026	
Date	Document Version	Document Revision History	Document Author/Reviser
10-09-2026	1.0	Initial BRD Draft Creation	Business Analyst
15-09-2026	1.1	Stakeholder & Technical Review	Lead Systems Architect

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1. Context
2. Business Justification
3. Functional and Non-functional Requirements

Task 2: AS-IS Process Map

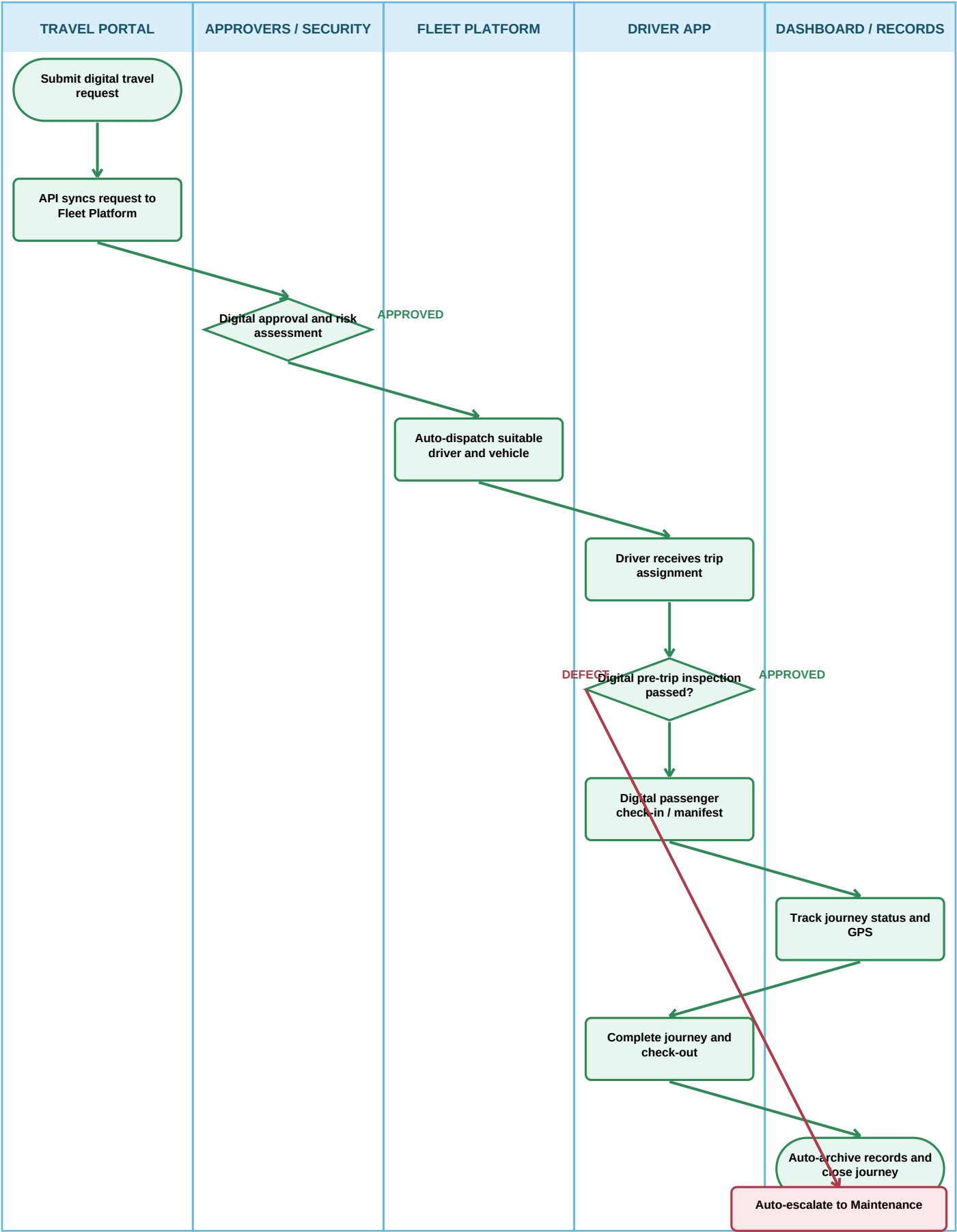
Current travel and fleet coordination workflow - manual hand-offs and paper controls



Legend: **Oval** = Start / End **Rectangle** = Process **Diamond** = Decision **Document** = Paper / Record

Task 3: TO-BE Process Map

Integrated future-state workflow - automated approvals, dispatch, monitoring and archiving



Legend: Oval = Start / End Rectangle = Process Diamond = Decision Document = Paper / Record

4. Document Approval

1. Context

1.1 Background and Problem Statement

Energy Company manages natural gas, oil, and fuel distribution from a centralized island facility. As operational travel increases between headquarters and field sites, the current process has become difficult to manage because employees, line managers, security teams, approvers, fleet managers, and drivers rely on manual coordination. The absence of integration between the Travel Management Portal and fleet operations creates delays, duplicate paperwork, fragmented security assessments, manual vehicle inspection records, and paper-based passenger manifests.

1.2 Current State (As-Is) & Future State (To-Be) Process Summary

As-Is: The current process follows a fragmented 11-step workflow that depends on separate manual forms, paper passenger manifests, hand-signed inspection checklists, and disconnected communication channels. This increases processing time and limits visibility across the full journey lifecycle.

To-Be: The proposed Fleet Management Platform will provide an integrated end-to-end process with Travel Portal API synchronization, digital approvals, automated driver and vehicle dispatching, mobile inspection and passenger manifest capture, and real-time dashboard monitoring.

2. Business Justification

2.1 Expected Benefits

Benefit Type	Expected Benefit
Qualitative Benefits	Improves the employee travel experience, strengthens driver and passenger safety through digital pre-trip inspections, removes the need for physical paper storage, and creates a complete audit trail from request submission to journey closure.
Quantitative Benefits	Targets a 60% reduction in journey processing cycle time, a 90% reduction in paperwork handling, 100% completion of pre-journey safety checks, and better fleet utilization through improved assignment visibility.

2.2 Stakeholder Mapping / Beneficiaries

Stakeholder / Beneficiary	How They Benefit
Traveling Employees	Submit official travel requests through a single digital process without resubmitting manual forms to Logistics.
Line Managers and Signatories	Review, approve, or reject travel requests quickly through a structured digital approval workflow.
Security Team	Complete digital risk assessments, record destination accessibility, and maintain centralized security approval records.
Fleet Managers	Match drivers and vehicles more efficiently while maintaining real-time visibility of fleet availability and journey status.
Drivers	Receive trip assignments, complete vehicle inspections, and manage passenger manifests through a mobile application.

3. Functional and Non-Functional Requirements

3.1 Functional Requirements

S/N	Functional Requirement	Priority	Comments
1	Travel Portal Integration: The platform must automatically receive approved travel request data from the Travel Management Portal via API, including employee ID, travel dates, destination, and journey details.	M	Removes the need for manual journey schedule forms.
2	Automated Workflow Routing: The platform must route requests sequentially to the line manager, Security Team, and required signatories for digital approval.	M	Supports automated email or app notifications for pending approvals.
3	Digital Security Assessment: The Security Team must be able to record destination accessibility, risk ratings, approval decisions, and escalation notes in the platform.	M	Creates one central security risk record for each journey.
4	Smart Fleet Dispatch: The platform must recommend or assign available drivers and suitable vehicles based on journey requirements, driver availability, and vehicle status.	M	Reduces manual planning effort for the Fleet Manager.

5	Mobile Driver Vehicle Inspection: Drivers must be able to complete digital pre-trip safety checklists on a mobile device before departure.	M	Automatically flags failed checks or defects to the fleet team.
6	Digital Passenger Manifest: Passengers must be checked in and confirmed through a mobile digital process, such as QR code scanning or app-based verification.	M	Replaces paper manifests and improves passenger accountability.
7	End-to-End Journey Dashboard: The platform must provide a central dashboard showing request status, approval progress, dispatch status, journey progress, inspection outcomes, and completion status.	M	Enables real-time monitoring and management reporting.
8	Automated Fleet Maintenance Alerts: The platform should generate maintenance work orders when a vehicle fails a pre-trip inspection or when a defect is reported.	D	Improves maintenance follow-up and vehicle readiness.

3.2 Non-functional Requirement (NFR)

S/N	NFR Category	Requirement
1	Performance	The platform must respond within 2 seconds for standard portal requests, searches, and dashboard views under normal operating conditions.
2	Availability and Reliability	The platform must support 99.9% availability to enable continuous operations across headquarters and field locations.
3	Security and Compliance	The platform must support role-based access control, secure Active Directory authentication, and encryption of sensitive passenger and journey data.
4	Usability and Mobility	The mobile app must be simple to use and must support offline data capture for inspections and journey activities in low-connectivity areas.

4. Document Approval

S/NO	NAME	ROLE	DEPARTMENT	SIGNATURE	DATE
1.	Operations Head	Requester	Fleet & Logistics	Digital Sign	10-09-2026
2.	Managing Director	Sponsor (GH)	Executive Management	Digital Sign	10-09-2026

3.	Lead Architect	Enterprise Architect (EA)	IT & Systems	Digital Sign	10-09-2026
4.	Lead Engineer	Developer	Engineering	Digital Sign	10-09-2026
5.	Senior BA	Business Relationship Manager (BRM)	Business Analysis	Digital Sign	10-09-2026

Approval by Chief Data Officer (CDO)

Status: [X] Approved [] Declined

Comments: Approved to proceed to the technical build and system integration phase.

CDO's Signature: O.C.C

Date: 10-09-2026

5. MoSCoW Application

Energy Company Fleet Management Platform

The MoSCoW framework prioritises the requirements identified in the Capstone Project and Business Requirements Document (BRD).

MoSCoW Category	Requirement	Project / BRD Context
Must Have	Travel Portal API Integration	Automatically sync approved travel requests into the Fleet Platform, replacing paper and manual journey forms.
Must Have	Automated Approval Routing	Route requests sequentially to Line Managers, Security, and Signatories for digital sign-off.
Must Have	Digital Security Risk Assessment	Record destination accessibility, risk ratings, approval decisions, and centralised security records.
Must Have	Smart Fleet Dispatch Engine	Match suitable drivers and vehicles using availability, licence type, vehicle status, and journey parameters.
Must Have	Mobile Driver Pre-Trip Inspection	Provide a mobile safety checklist for drivers to complete before vehicle departure.
Must Have	Digital Passenger Manifest	Enable passenger check-in and check-out through QR code or app verification, replacing paper manifests.
Must Have	End-to-End Central Dashboard	Provide real-time visibility across request status, approvals, dispatch, inspections, and journey execution.
Should Have	Automated Maintenance Work Orders	Trigger maintenance alerts and work orders when pre-trip defects or failed safety checks are recorded.
Could Have	Offline Mobile Data Capture	Capture inspections and manifest updates in low-connectivity areas and synchronise when reconnected.
Won't Have (Initial Release)	Advanced Predictive Fleet Analytics	Reserve machine-learning failure prediction and automated route optimisation for future post-deployment phases.